

23 : 1st half.13-AM(aa)

Con. 9152-13.**GS-6420**

(3 Hours)

[Total Marks : 100]

- N.B. :** (1) Question No. 1 is **compulsory**.
 (2) Solve any **four** out of the remaining **six** questions.
 (3) Draw neat **diagram** whenever **necessary**.

1. (a) Convert $(1234.56)_{10}$ to octal, hexadecimal. 4
- (b) Represent $(29)_{10}$ into XS-3 and Gray code. 4
- (c) Design full adder using half adders. 4
- (d) Design a full subtractor using a decoder and additional gates. 4
- (e) Simplify and implement using gates 4

$$Y = \overline{AB} (B + C) + AB (\overline{B + C})$$

2. (a) What is canonical SOP and POS form ? Explain with an example. 5
- (b) Implement the following using only one 8 : 1 MUX and few gate 5

$$F(A, B, C, D) = \sum m(0, 3, 5, 7, 9, 13, 15)$$

- (c) Using the k-map method minimization technique simplify and draw the circuit for the following function, 10

$$F(A, B, C, D, E) = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 11, 14, 16, 17, 18, 19) + d(24, 25)$$

3. (a) Simplify using Quine Mc-Cluskey method. Realize the equation using any universal gate. 10

$$F(A, B, C, D) = \pi M(0, 2, 3, 6, 7, 8, 9, 12, 13)$$

- (b) Design a BCD adder using 4 bit binary adders and explain. 10

4. (a) Design a 3 bit even and odd parity generator. 10
- (b) State truth table of 3 bit gray to binary conversion and design using 3 : 8 decoder and additional gates. 10

5. (a) Design MOD - 6 synchronous counter and explain its operation. 10
- (b) Draw a 4 bit universal shift register and explain. 10

6. (a) Design 4 bit ring counter using J K ff, draw the timing diagram for the same. 10
- (b) Compare TTL and CMOS logic families. 10

7. Write notes on the following (any **four**) :— 20

- (a) ALU (Arithmetic Logic Unit)
 - (b) PAL and PLA
 - (c) Race around condition
 - (d) Error detecting and correcting code
 - (e) Applications of flip flops and registers.
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